1 ECE 115C Notes

1.1 MOS Transistor

1.1.1 Channel Length Modulation

V_{min} determines the operating region.

$$V_{min} = min\{V_{GT},\ V_{DS},\ V_{DSSAT}\}$$

Entering Saturation, Linear, or V-SAT respectively. Here $V_{GT} = V_{GS} - V_T$, $k'_n = \mu_n C_{ox}$, and $k_n = K'_n W/L$. Early Effect λ does not exist in linear.

$$I_D = I'_D(1 + \lambda V_{DS})$$
$$I_D = k'_n \frac{W}{L}(V_{GT}V_{min} - \frac{V_{min}^2}{2})(1 + \lambda V_{DS})$$

Subthreshold Current

$$I_D = I_0 \exp \frac{V_{GS}}{b\Phi_t}(1 - \exp(-\frac{V_{DS}}{\Phi_t})),\ \Phi_t = \frac{kt}{q}$$

1.1.2 RC Model

Resistance R_{on} when active

$$R_{on} \approx \frac{3}{4}\frac{V_{DD}}{I_{DSAT}}\left(1 - \frac{5}{6}\lambda V_{DD}\right) = \frac{3}{4}\frac{V_{DD}}{I_{DSAT}}$$

Capacitance, Let $C_g = C_{ox}WL_{eff}$

	C_{GCB}	C_{GCS}	C_{GCD}
Cutoff	C_g	0	0
Triode	0	$C_g/2$	$C_g/2$
Satura	0	$2/3 \times C_g$	0

Cap at output is called C_{par}, at gate is called C_{gate}, so $\gamma = C_{par}/C_{gate}$

1.1.3 Miller's

Impedance z_F connecting two side of an amplifier may be converted to two grounded impedance z_1 and z_2.

$$z_1 = \frac{z_F}{1 - A} \qquad z_2 = \frac{z_F}{1 - A^{-1}}$$

1.1.4 Power

Dynamic

1. Energy $E = C_L V_{DD} V_{swing}$
   (a) $E_{0\to1} = E_{heat} + E_c$,
   $E_{heat} = E_{0\to1} - E_c$ $E_c = 0.5C_L V^2$
   (b) $E_{1\to0} = 0$, $E_{heat} = E_c$
2. Power $P = \alpha_{0\to1} f_{clk} E$

Short Circuit

1. Energy $E = t_{sc} V_{DD} I_{peak}$
   $t_{sc} = t_{slope}/0.8 \times (V_{DD} - 2V_T)/V_{DD}$
2. Power $P = \alpha_{0\to1} f_{clk} E$

Leakage Current

$$I_{DS} = I_0 \frac{w}{w_0} 10^{((V_{GT} - \gamma V_{DS})/s)}$$

Refer to CMOS logic for α.

1.2 CMOS Logic

1.2.1 Switching Probability

Output x can be 1 or 0, $P(x = 0) + P(x = 1) = 1$, so $\alpha_{x:0\to1} = P(x = 0)P(x = 1)$

1.2.2 Gate Sizing

Find reference inverter, R_{on} needs to be matched, consider an original size 1, n in series means each of size n, $R_{on} = R/n$, n in parallel size unchanged.

1.2.3 Propagation Delay

Delay from τ and HL transition

$$t = ln(2)\tau \qquad t_{pHL} = \ln(2) C_L R_{on}$$

Determining the worst case., if Low to High (or High to Low), we Propagation Delay. Each Stage of Gates is isolated by C_g, so cut all inputs, and calculate τ for each stage using RC (R from pull-up or down resistance, C from C_p and next stage C_g).

$$t_p \approx \ln(2) R_g C_{in,\ gate}\left(\frac{C_{par,\ gate}}{C_{in,\ gate}} + \frac{C_{out}}{C_{in,\ gate}}\right)$$

Unitless Delay (d) is the logical effort (g) + parasitic effort (p)

$$d = \frac{\tau_p}{\tau_{inverter}} = \frac{\tau_{gate}}{\tau_{inv}}(\gamma + h)$$
$$= gh + p\gamma = g(\gamma_{gate} + h) = \tau(p + gh)$$

Logical effort is the ratio of C_{gate} to which of the reference inverter (R_{on} needs to be matched, so only scale this to the unit inverter and nothing more should be done)

$$g = \frac{C_{in,gate}}{C_{in,inv}} = \frac{\tau_{gate}}{\tau_{inv}}$$
$$= \frac{R_{gate}C_{gate}}{R_{inv}C_{inv}}$$

Parasitic Effort is the ratio of the output C_{par} to the reference inverter (NAND and NOR both p = 2)

$$p = \frac{C_{par,gate}}{C_{par,inv}} = g\gamma_{gate}$$

Here Cin and Cout should consider only the gate not the external factor unless appended between stage, and will have to be modeled as part of the inner gate.

1.2.4 Multistage

All stages should have the same logical effort for minimum delay. Fanout factor $f = gh$, Total fanout $F = f^N = C_L/C_{in,1}$,

$$f_{opt} = \exp(1 + \gamma/f)$$

Branching Effort

$$b = \frac{C_{on-path} + C_{off-path}}{C_{on-path}}$$

Path Electrical Effort H, Logical Effort G, Branching Effort B, and Path Effort F

$$\begin{aligned} H =& C_{out}/C_{in} = h_1 \times h_2 \cdots \times h_n \\ G =& g_1 \times g_2 \cdots \times g_n \\ B =& b_1 \times b_2 \cdots \times b_n \\ F =& GBH = f^n \end{aligned}$$

Total Delay D, can mostly assume $\gamma = 1$ in course context

$$D = N(f + \gamma) = \sum_{stages} \tau(p + gh)$$

To optimize D in terms of sizes, write out as capacitance and partial differentiate.

1.3 Wire

Wire Modeling (May provide sheet resistance, check unit first)

$$R = \frac{\rho L}{A}$$

Elmore Delay $t = 0.69\tau$, consider τ for each capacitor and the Resistors that are in the path to vout before it. And for driving an RC-Line

$$t_p = 0.69 R_s C_w + 0.38 R_w C_w$$

1.4 Adder

For a N bit M stage adder, each stage with a propagation logic

$$\begin{aligned} t_{adder} =& t_{setup} + M t_{carry,p} + \left(\frac{N}{M} - 1\right) t_{Mux} \\ &+ (M-1) t_{carry} + t_{sum} \end{aligned}$$

1.5 Timing

In circuit there are delays and randomness impacting the clock. Delays t_{skew} denoted as δ which may be positive (clk with In) or negative (clk closer to In). Randomness can be $\pm t_{jitter}$. Negative skew improves race margin. Jitter almost never changes but the sign of skew needs adjustments.

$$t_{cu} = \delta + 2t_{jitter}$$

Max Skew

$$t_{skew,max} = t_{c-q,cd} + t_{logic,min} - t_{hold}$$

Max Jitter

$$t_{Jitter,max} = t_{c-q,min} + t_{pd,min} - t_{hold} - t_{skew}$$

Adjust skew of sign according to its sides. Clock cycle time must satisfy

$$\begin{aligned} T_{clk} \geq& t_{c-q} + t_{logic} + t_{su} + t_{cu} \\ T_{clk} \geq& t_{c-q,max} + t_{logic,min} + t_{su} - \delta \end{aligned}$$

Race Margin must satisfy, cause race problem otherwise.

$$\begin{aligned} t_{hold} \leq& t_{c-q,cd} + t_{logic,cd} - t_{cu} \\ t_{hold} \leq& t_{c-q,cd} + t_{logic,cd} - \delta \end{aligned}$$

(pd means logic, cd means minimum)